

# Kvantemekanik Formelsamling

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## Indhold

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## 1 Electrostatics

### 1.1 Electric field

The force on a test charge ( $Q$ ) from a single point charge ( $q$ ), a distance  $\bar{r}$  away, is calculated by Coulomb's law:

$$\bar{F} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \hat{r}, \quad (2.1)$$

where  $\epsilon_0 = 8.85 \cdot 10^{-12} \frac{\text{C}^2}{\text{N}\cdot\text{m}^2}$  is the permittivity of free space.

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